## Experiment: Patient Database, Schema Validation, and Indexing in MongoDB

### a. Create Database and Collections with Validation

```
use hospitalDB

db.createCollection("patients", { validator: { $jsonSchema: { bsonType:
"object", required: ["patientId","name","age","dateOfBirth"], properties: {
patientId: { bsonType: "int" }, name: { bsonType: "string" }, age: {
bsonType: "int", minimum: 0, maximum: 120 }, dateOfBirth: { bsonType:
"date" } } } } })

db.createCollection("doctors", { validator: { $jsonSchema: { bsonType:
"object", required: ["doctorId","name","specialization"], properties: {
doctorId: { bsonType: "int" }, name: { bsonType: "string" },
specialization: { bsonType: "string" } } } } })

db.createCollection("medicines", { validator: { $jsonSchema: { bsonType:
"object", required: ["medicineId","name","category","price","expiryDate"],
properties: { medicineId: { bsonType: "int" }, name: { bsonType: "string"
}, category: { bsonType: "string" }, price: { bsonType: "double" },
expiryDate: { bsonType: "date" } } } } })

db.createCollection("prescriptions", { validator: { $jsonSchema: {
bsonType: "object", required:
["prescriptionId","patientId","doctorId","medicineId","dateIssued"],
properties: { prescriptionId: { bsonType: "int" }, patientId: { bsonType:
"int" }, doctorId: { bsonType: "int" }, medicineId: { bsonType: "int" },
dateIssued: { bsonType: "date" } } } } })

db.createCollection("sales", { validator: { $jsonSchema: { bsonType:
"object", required: ["saleId","medicineId","quantity","saleDate"],
properties: { saleId: { bsonType: "int" }, medicineId: { bsonType: "int" },
quantity: { bsonType: "int" }, saleDate: { bsonType: "date" } } } } })
```

### b. Insert Sample Data (10 Documents per Collection)

```
db.patients.insertMany([
{patientId:1,name:"Alice Johnson",age:29,dateOfBirth:new Date("1996-03-
15")},
{patientId:2,name:"Bob Smith",age:45,dateOfBirth:new Date("1980-07-22")},
{patientId:3,name:"Charlie Lee",age:60,dateOfBirth:new Date("1965-01-10")},
{patientId:4,name:"David Kim",age:34,dateOfBirth:new Date("1991-11-03")},
{patientId:5,name:"Eva Brown",age:51,dateOfBirth:new Date("1974-05-09")},
{patientId:6,name:"Frank Green",age:72,dateOfBirth:new Date("1953-02-14")},
{patientId:7,name:"Grace White",age:40,dateOfBirth:new Date("1985-06-17")},
{patientId:8,name:"Henry Miller",age:25,dateOfBirth:new Date("2000-08-
20")},
{patientId:9,name:"Ivy Davis",age:37,dateOfBirth:new Date("1988-09-29")},
{patientId:10,name:"Jack Wilson",age:55,dateOfBirth:new Date("1970-04-05")}
])

db.doctors.insertMany([
{doctorId:1,name:"Dr. Emily Carter",specialization:"Cardiology"},
{doctorId:2,name:"Dr. Michael Brown",specialization:"Neurology"},
{doctorId:3,name:"Dr. Sophia Lee",specialization:"Pediatrics"},
{doctorId:4,name:"Dr. David Chen",specialization:"Orthopedics"},
{doctorId:5,name:"Dr. Laura White",specialization:"Dermatology"},
{doctorId:6,name:"Dr. James Wilson",specialization:"Psychiatry"},
{doctorId:7,name:"Dr. Olivia Smith",specialization:"Gynecology"},
```

```
{doctorId:8,name:"Dr. Robert Clark",specialization:"Oncology"},
{doctorId:9,name:"Dr. Linda Johnson",specialization:"Ophthalmology"},
{doctorId:10,name:"Dr. William Scott",specialization:"ENT"}
])

db.medicines.insertMany([
{medicineId:1,name:"Paracetamol",category:"Pain
Relief",price:2.5,expiryDate:new Date("2026-05-01")},
{medicineId:2,name:"Amoxicillin",category:"Antibiotic",price:5.0,expiryDate
:new Date("2025-08-15")},
{medicineId:3,name:"Cetirizine",category:"Antihistamine",price:1.5,expiryDa
te:new Date("2027-01-10")},
{medicineId:4,name:"Ibuprofen",category:"Pain
Relief",price:3.0,expiryDate:new Date("2026-10-20")},
{medicineId:5,name:"Metformin",category:"Diabetes",price:4.0,expiryDate:new
Date("2025-12-05")},
{medicineId:6,name:"Aspirin",category:"Blood
Thinner",price:2.0,expiryDate:new Date("2027-03-12")},
{medicineId:7,name:"Atorvastatin",category:"Cholesterol",price:6.0,expiryDa
te:new Date("2025-07-30")},
{medicineId:8,name:"Omeprazole",category:"Acidity",price:3.5,expiryDate:new
Date("2026-09-25")},
{medicineId:9,name:"Salbutamol",category:"Asthma",price:7.0,expiryDate:new
Date("2025-11-11")},
{medicineId:10,name:"Vitamin
D",category:"Supplement",price:4.5,expiryDate:new Date("2027-02-19")}
])

db.prescriptions.insertMany([
{prescriptionId:1,patientId:1,doctorId:1,medicineId:1,dateIssued:new
Date("2025-08-01")},
{prescriptionId:2,patientId:2,doctorId:2,medicineId:2,dateIssued:new
Date("2025-08-02")},
{prescriptionId:3,patientId:3,doctorId:3,medicineId:3,dateIssued:new
Date("2025-08-03")},
{prescriptionId:4,patientId:4,doctorId:4,medicineId:4,dateIssued:new
Date("2025-08-04")},
{prescriptionId:5,patientId:5,doctorId:5,medicineId:5,dateIssued:new
Date("2025-08-05")},
{prescriptionId:6,patientId:6,doctorId:6,medicineId:6,dateIssued:new
Date("2025-08-06")},
{prescriptionId:7,patientId:7,doctorId:7,medicineId:7,dateIssued:new
Date("2025-08-07")},
{prescriptionId:8,patientId:8,doctorId:8,medicineId:8,dateIssued:new
Date("2025-08-08")},
{prescriptionId:9,patientId:9,doctorId:9,medicineId:9,dateIssued:new
Date("2025-08-09")},
{prescriptionId:10,patientId:10,doctorId:10,medicineId:10,dateIssued:new
Date("2025-08-10")}
])

db.sales.insertMany([
{saleId:1,medicineId:1,quantity:5,saleDate:new Date("2025-08-01")},
{saleId:2,medicineId:2,quantity:3,saleDate:new Date("2025-08-02")},
{saleId:3,medicineId:3,quantity:2,saleDate:new Date("2025-08-03")},
{saleId:4,medicineId:4,quantity:4,saleDate:new Date("2025-08-04")},
{saleId:5,medicineId:5,quantity:1,saleDate:new Date("2025-08-05")},
{saleId:6,medicineId:6,quantity:6,saleDate:new Date("2025-08-06")},
{saleId:7,medicineId:7,quantity:3,saleDate:new Date("2025-08-07")},
{saleId:8,medicineId:8,quantity:2,saleDate:new Date("2025-08-08")},
{saleId:9,medicineId:9,quantity:5,saleDate:new Date("2025-08-09")},
```

```
{saleId:10,medicineId:10,quantity:4,saleDate:new Date("2025-08-10")}
])
```

### c. Creating Indexes

### i. Single Field Index

```
db.patients.createIndex({patientId:1})
```

### ii. Compound Index

```
db.prescriptions.createIndex({patientId:1,dateIssued:-1})
```

### iii. Text Index

```
db.medicines.createIndex({name:"text",category:"text"})
db.medicines.find({$text:{$search:"pain"}})
```

### iv. Partial Index

```
db.medicines.createIndex({expiryDate:1},{partialFilterExpression:{expiryDat
e:{$gte:new Date()}}})
```

### v. Unique Index

```
db.doctors.createIndex({doctorId:1},{unique:true})
```

### d. Example Queries Using Indexes

```
// Single field index
db.patients.find({patientId:5})

// Compound index
db.prescriptions.find({patientId:3}).sort({dateIssued:-1})

// Text index
db.medicines.find({$text:{$search:"antibiotic"}})

// Partial index
db.medicines.find({expiryDate:{$gte:new Date()}})

// Unique index: duplicate doctorId will fail
db.doctors.insertOne({doctorId:1,name:"Duplicate",specialization:"Test"})
```

### e. Index Verification

```
db.medicines.getIndexes()
db.patients.getIndexes()
db.prescriptions.getIndexes()
db.doctors.getIndexes()
```

### f. Performance Comparison Example

### i. Without Index

```
db.patients.dropIndexes()
db.patients.find({patientId:7}).explain("executionStats")
```

### ii. With Index

```
db.patients.createIndex({patientId:1})
db.patients.find({patientId:7}).explain("executionStats")
```

**Result:** Schema validation ensures that documents follow the specified rules. Indexes improve query efficiency, text indexes support keyword searches, partial indexes index only matching documents, and unique indexes prevent duplicate indexed values. The `explain("executionStats")` output can be used to compare query execution statistics before and after indexing.